GROUP PROJECT SUBMISSION FORM

PROGRAMME: BSDA

SEMESTER: April 2025

SUBJECT: BIS3106 Project Management

DEADLINE:

PROJECT HANDED IN BY 14 Week
PRESENTATION: Week 13 (Tutorial 1-5 sessions, Tue & Wed)

INSTRUCTIONS TO CANDIDATES

- This is a group project with no presentation requirement.

IMPORTANT
The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

- Coursework submitted after the deadline but within 1 week will be accepted for a maximum mark of 50%.
- Work handed in following the extension of 1 week after the original deadline will be regarded as a non-submission and marked zero.

Lecturer's Remark (Use additional sheet if required)
List down the names of the group members and the student IDs here.

I Isabelle Yau Li Shuang (23021165), Peggy Liow En Ki (23020852), Leow Yan Yi (23020324), Ooi Shi Qi (21098272) received the assignment and read the comments.

Isabelle, Peggy, Yan Yi, Shi Qi (30/07/2025) (Signature/Date)

Academic Honesty Acknowledgement

"I Isabelle Yau Li Shuang, Peggy Liow En Ki, Leow Yan Yi, Ooi Shi Qi (Student's Name) verify that this paper contains entirely my own work. I have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, I have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. I realize the penalties *(refer to page 16, 5.5, Appendix 2, page 44 of the student handbook diploma and undergraduate programme)* for any kind of copying or collaboration on any assignment."

Isabelle, Peggy, Yan Yi, Shi Qi (30/07/2025) (Student's signature / Date)

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Table of Contents

1.0 Introduction **2**
2.0 Project Charter **7**
3.0 Project Scheduling **13**
3.1 Milestone 13
3.2 Gantt Chart 16
3.3 Work Breakdown Structure 17
4.0 Project Resource **19**
5.0 Project Risk **22**
6.0 Project Quality **25**
7.0 Project Closure **27**
7.1 Sample Project Close-Out Meeting Agenda 28
7.2 Lesson Learned Template 29
8.0 References **30**

1.0 Introduction

Our project title is "Malaysia Airlines System Enhancement Project", which ran from 12th May 2025 to 29th July 2025, with a duration of 11 weeks. This project aims to improve the overall user experience, as the current Malaysia Airlines app has several usability issues, particularly the manual check-in process and unspecified error messages that reduce user satisfaction. It will also enhance customer support accessibility by introducing a 24-hour live chat feature with AI-assisted capabilities, addressing the current lack of in-app customer support. Besides, the project will improve login accessibility for travellers by adding email-based OTPs as an alternative verification method, resolving the inconvenience of depending on SMS or WhatsApp authentication.

The objectives of the project are:

- To improve the check-in experience by allowing users to select upcoming flights directly from a list at the check-in page, eliminating the need to enter booking details manually.
- To improve customer support accessibility by integrating a 24-hour in-app live chat feature, including an AI-powered chatbot, to provide immediate assistance.
- To enhance the login flexibility by introducing an additional verification method using email-based OTPs to make the login process more accessible for international travellers and eSIM users.
- To improve the error message clarity by replacing unspecified error messages with specific prompts that help users to understand and resolve issues more effectively.

The deliverables of the project are:

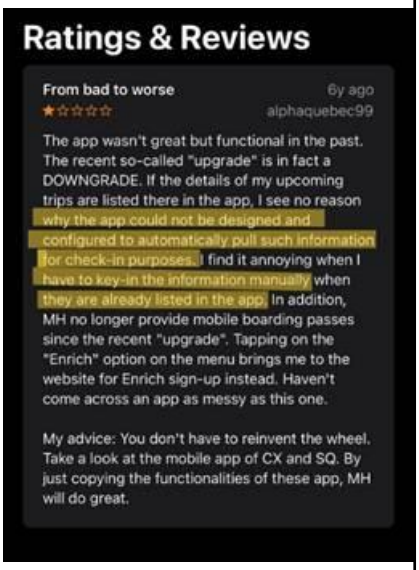
- A redesigned check-in page that shows a list of upcoming flights linked to the user's account.
- An integrated 24-hour in-app live chat system with human agent support, where an AI-powered chatbot handles common queries. Backend integration with user account data enables personalized support.
- An improved login experience by adding email-based OTP as an alternative two-factor authentication method.
- An improved error message system with contextual error messages throughout the app, especially on the "Manage Flight" page with suggested user actions to help resolve issues easily.

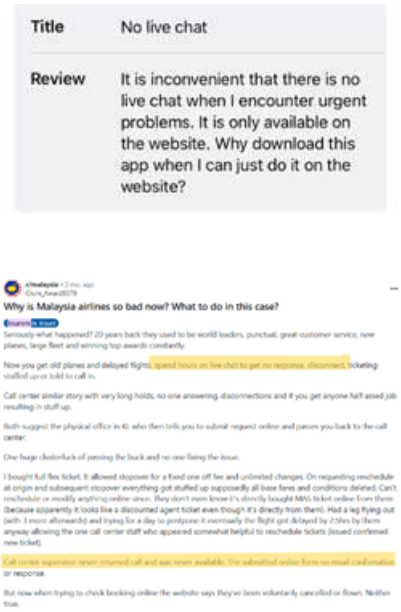
One of the components included in the scope of this project is the redesign of the check-in interface to display a list of upcoming flights for easier user access. We will also include a 24-hour in-app live chat system and AI-powered chatbot customer support into the Malaysia Airlines mobile application. Besides, an email-based OTP will be added as an alternative login verification method and error messages will be enhanced with clear explanations and suggested user actions, especially on the "Manage Flight" page. Functional testing and user acceptance testing will be carried out for all enhanced features to ensure they meet business needs. However, this project will not cover enhancements to the Malaysia Airlines web version, as we focus on the mobile application. It also does not include a full redesign of the app's UI/UX interface, except for the check-in, login and manage flight pages. The marketing of updated app features and the development of unrelated functionalities such as loyalty programs will not be included in this project.

The key stakeholders included in this project are the project sponsor, the project manager, the infrastructure engineer, the business analyst and the developer. The project sponsor is Dr. Abdul Aziz Abdullah, who is responsible for the overall accountability of the project. He reviews and approves the final deliverables such as redesigned check-in interface, live chat system, email OTP feature and enhanced error messaging. Next, the project manager of the team is Isabelle Yau Li Shuang. She is responsible for communicating with stakeholders, managing all project resources and ensuring the final deliverables are completed, tested and delivered within the 11-week timeframe. Moreover, Ooi Shi Qi is the business analyst of the project, who is responsible for gathering and analysing the project's requirements. She works with stakeholders to identify the problems with the current app such as manual check-in issues and defines what improvements are needed. She is also responsible for carrying out User Acceptance Testing, by preparing test scenarios, supporting testing efforts and verifying that the enhanced features function as intended. Furthermore, Leow Yan Yi is the infrastructure engineer of the team. She is responsible for maintaining the system's architecture, integration and security to ensure stability and safeguard data protection. She prepares the backend infrastructure to support new features such as live chat, email OTP and enhanced error handling without causing system downtime. She also ensures that sensitive data is stored and transmitted in compliance with data protection standards. Finally, the developer of the project team is Peggy Liow En Ki. She is responsible for implementing new features such as developing and integrating the redesigned check-in interface which displays a list of upcoming flights. She also resolves bugs found during testing or after deployment, especially those affecting the new features.

There are a few constraints included in this project. Firstly, all design, development, testing and deployment tasks must be completed within 11 weeks, which limits the possibility of delays or scope expansion. Besides, enhancements are limited to the mobile app, particularly check-in, login, manage flight and live chat customer support features. No enhancements will be performed on the web version of Malaysia Airlines or other modules like booking or loyalty programs. Moreover, delays might happen in releasing the updated Malaysia Airlines app as Apple App Store and Google Play must perform reviews before releasing it. All enhanced features must comply with the Personal Data Protection Act 2010 and the General Data Protection Regulation laws, especially for handling personal data during live chat and OTP verification.

One of the assumptions made in this project is that users are assumed to already have Malaysia Airlines accounts which includes upcoming flights for the check-in feature to work. It is also assumed that users' email addresses in the system are valid and verified to successfully receive OTPs during the login process. Furthermore, the development team assumes that they can access user flight information and booking data through Malaysia Airlines' backend systems. The technical environment of the Malaysia Airlines app is also assumed to allow integration with third-party or custom live chat.

Issue	**Screenshots of the review extracted from the App Store**	**Suggested Mitigation**	**Justification**
Issue#1: The Malaysia Airlines app has an unfriendly check-in system, requiring users to manually enter their booking reference and last name, rather than simply selecting	 Ratings & Reviews From bad to worse 6y ago ★☆☆☆☆ alphaquebec99 The app wasn't great but functional in the past. The recent so-called "upgrade" is in fact a DOWNGRADE. If the details of my upcoming trips are listed there in the app, I see no reason why the app could not be designed and configured to automatically pull such information for check-in purposes. I find it annoying when I have to key-in the information manually when they are already listed in the app. In addition, MH no longer provide mobile boarding passes since the recent "upgrade". Tapping on the "Enrich" option on the menu brings me to the website for Enrich sign-up instead. Haven't come across an app as messy as this one. My advice: You don't have to reinvent the wheel. Take a look at the mobile app of CX and SQ. By just copying the functionalities of these app, MH will do great.	We suggest displaying a list of upcoming flights on the check-in page, allowing users to directly select the flight they wish to check-in for.	Users don't have to check back their email and search for the booking confirmation to retrieve their booking reference. They can just simply select their flight from the displayed list.

their flight from a list.			
Issue#2: The Malaysia Airlines app lacks in-app customer support via live chat, making it difficult for users to get immediate assistance when needed.	 Title: No live chat Review: It is inconvenient that there is no live chat when I encounter urgent problems. It is only available on the website. Why download this app when I can just do it on the website? Why is Malaysia airlines so bad now? What to do in this case?	We suggest integrating a 24-hours live chat feature within the Malaysia Airlines app, which allows users to connect with customer support agents in real-time. An AI-powered chatbot can be implemented to handle common inquiries and escalate complex issues to human agents when necessary.	Currently, the Malaysia Airlines app only provides customer service through a 24-hour general helpline. If live chat feature is integrated, the system can directly retrieve users' flight information from their accounts without the need for users to provide booking details verbally. AI-powered chatbot improves efficiency by instantly addressing frequently asked questions.

Issue#3: The two-factor verification during Malaysia Airlines app login is inconvenient, because it relies on SMS codes. Users must switch to their local SIM card to receive the verification message, which is especially problematic for travellers using international or eSIM services.	**Ratings & Reviews** Room to improve — 9 Feb ★☆☆☆☆ wafiqfarhan 1. Payment can be much better and faster. Booking hotel room is much faster 2. Flight booking is not easy especially when you want to add baby name with your name. 3. Please remove the cookies policy. When open any promotion, it pop up the cookies which cannot click any button because phone not supported and im using iphone. 4. Jounify, going places, temp, etc not going to work with phone apps. Please rectify this issue. 5. Sign verification - please remove sms for verification for each time login. Other app EK or QR didnt use this, so why so secure with dual security and all. If im purchasing ticket from outside country, i need to open my local sim to retrieve this sms. 6. why payment in different currency etc. Let only payment in the currency selected in the setting. 7. Check- in feature is not friendly. Thanks for auto add trip in mytrip tab. But checkin suppose to be integrated. **Useless app when you're in oversea** ★☆☆☆☆ 2y ago · Emeerazhar When you're abroad, you'll change your simcard for internet. Now I cannot login to the app because the verification only can be done through sms code. Malaysia airlines should do several verification options. Please learn from google! Stupid	We suggest adding an additional verification method on the login page by allowing OTP to be sent to the user's email.	Relying solely on SMS and WhatsApp for two-factor verification can be problematic for users who are traveling internationally or using an eSIM without access to their local number. By adding email as an alternative method for receiving OTPs, users can log in securely without needing to switch SIM cards.
Issue#4: The Malaysia Airlines app frequently displays unspecified error messages on the "Manage flight" page, leaving users unsure about what went wrong or how to resolve the issue.	A Google user September 12, 2018 The manage flight and add services error replies are meaningless. When checked in using pc this apps does not show updated data for seats booked etc. It will still allow check in an seat bookings. Apps need to have meaningful error messages and must reflect actions tak on the website for manage bookings 13 people found this review helpful Did you find this helpful? Yes No Malaysia Aviation Group — September 18, 2018 Hi Ye, thank you for your feedback. Can we request you to send us more details of these error details at digital.solutions@malaysiaairlines.com in order to review it further.	We suggest displaying specific error messages that explain the issue and suggest possible actions.	Clear error messages help users understand the problem and how to fix it.

2.0 Project Charter

Project Title	Malaysia Airlines System Enhancement Project		**Project Manager**	Isabelle Yau Li Shuang	
Project Start Date	12/5/2025	**Project End Date**	29/7/2025	**Project Sponsor**	Dr Abdul Aziz Abdullah

Business Need

- To improve the check-in experience by allowing users to select upcoming flights directly from a list at the check-in page, eliminating the need to enter booking details manually.
- To improve customer support accessibility by integrating a 24-hour in-app live chat feature, including an AI-powered chatbot, to provide immediate assistance.
- To enhance the login flexibility by introducing an additional verification method using email-based OTPs to make the login process more accessible for international travellers and eSIM users.
- To improve the error message clarity by replacing unspecified error messages with specific prompts that help users to understand and resolve issues more effectively.

Project Scope

Within Scope:

- To redesign the check-in interface to display a list of upcoming flights for convenient user access.
- Implement a 24-hour in-app live chat feature, including an AI-powered chatbot to handle routine queries and escalate complex issues to human agents when needed.
- Introduce email-based OTP as an alternative method for two-factor authentication during login.
- Improve the clarity of error messages throughout the app, especially on the "Manage Flight" page by including specific prompts and recommended actions to resolve issues.

Outside of scope:

- Enhancements or changes to the Malaysia Airlines website are excluded, as this initiative focuses on the mobile application.
- A full redesign of the app's UI/UX interface is not included, only specific sections such as check-in, login and manage flight pages will be redesigned.

- Marketing or promotional efforts related to the new features will not be carried out in this project.
- The development of unrelated functionalities such as loyalty programs will not be included in this project.

Deliverables

- A redesigned check-in page that shows a list of upcoming flights linked to the user's account.
- An integrated 24-hour in-app live chat system with human agent support, where an AI-powered chatbot handles common queries. Backend integration with user account data enables personalized support.
- An improved login experience by adding email-based OTP as an alternative two-factor authentication method.
- An improved error message system with contextual error messages throughout the app, especially on the "Manage Flight" page with suggested user actions to help resolve issues easily.

Risk and Issues

1. In terms of user data privacy concerns, user's personal data like flight details, names and contact information will be involved when developing the AI and live chat customer support feature. If the data is not handled properly, it can pose data privacy risks which involves legal consequences and reputational damage of the company.
2. The project would exceed its allocated budget if the time required for testing, integration or UI development is underestimated during planning phase.
3. Compatibility issues may occur when we integrate new features such as live chat, AI chatbot, email OTP and enhanced check-in interface with Malaysia Airlines' existing backend systems.

Assumptions/ Dependencies

Assumptions:

- Users already have Malaysia Airlines accounts which includes upcoming flights for the check-in feature to work.
- Users' email addresses in the system are valid and verified to successfully receive OTPs during the login process.

- The development team assumes that they can access user flight information and booking data through Malaysia Airlines' backend systems.
- The technical environment of the Malaysia Airlines app is also assumed to allow integration with third-party or custom live chat.

Dependencies:

- The infrastructure and security teams must approve and support system changes, particularly for user authentication enhancements and API access.
- Email-based OTP functionality relies on users whether they have a valid and verified email addresses stored in the system.
- The project depends on timely access to Malaysia Airlines backend services that store user accounts, bookings and flight data for implementing features like upcoming flights on the check-in page.
- Design, development and testing activities depend on timely decision making and feedback from project sponsors and stakeholders

Financials

Cost Type	**Vendor/Labor Names**	**Rate (RM/ month)**	**Duration (months)**	**Amount (RM)**
Labor Costs	Project Manager	8000	2.5	20000
	Developer	6000	2.5	15000
	Business Analyst	6000	2.0	12000
	Infrastructure Engineer	7000	2.0	14000
Total Costs				61000

Milestones Schedule

Milestone	**Start Date**	**End Date**	**Expected duration**

Initiating phase			
1 Project Approval and Kick off Meeting	12/5/2025	12/5/2025	1 day
2. Identify problems in the current system and compare with other systems.	13/5/2025	13/5/2025	1 day
3. Identify the stakeholders and conduct stakeholder meeting.	14/5/2025	14/5/2025	1 day
4. Develop Project Charter	15/5/2025	15/5/2025	1 day
5. Identify Assumptions and Constraints	16/5/2025	17/5/2025	2 days
6. System Request	16/5/2025	17/5/2025	2 days
7. Prepare Business Case	18/5/2025	19/5/2025	2 days
8. Project Proposal	20/5/2025	20/5/2025	1 day
	Total days: 9 days		
Planning phase			
9. Background study	19/5/2025	20/5/2025	2 days
10. Feasibility study	21/5/2025	22/5/2025	2 days
11. Select Development Methodology	23/5/2025	24/5/2025	2 days
12. Determine Functional and Non-Functional Requirements	25/5/2025	26/5/2025	2 days
13. Review and Get Stakeholder Approval of Requirements	27/5/2025	27/5/2025	1 day
14. Develop Work Breakdown Structure (WBS)	28/5/2025	29/5/2025	2 days
15. Define Project Milestones and Deliverables	30/5/2025	31/5/2025	2 days
16. Assign Project Roles and Responsibilities (Job assignments)	1/6/2025	1/6/2025	1 day
17. Create Gantt Chart / Project Schedule	2/6/2025	4/6/2025	3 days
18. Develop Project Management Plan	2/6/2025	4/6/2025	3 days
19. Develop Risk Management Plan	2/6/2025	4/6/2025	3 days
20. Develop Communication Plan	2/6/2025	4/6/2025	3 days

21. Develop Procurement Plan	2/6/2025	4/6/2025	3 days
	Total days: 17 days		
3.0 Executing phase			
22. Direct and Manage Project Work	5/6/2025	7/6/2025	3 days
23. Acquire and Manage Project Team	8/6/2025	10/6/2025	3 days
24. Manage Communications	11/6/2025	13/6/2025	3 days
25. Integrate Planning with Execution	14/6/2025	16/6/2025	3 days
26. Define Backend Architecture and Data Integration Plan	17/6/2025	19/6/2025	3 days
27. Create Use Case Diagram and Entity Relationship Diagram (ERD)	20/6/2025	22/6/2025	3 days
28. Design Class Diagram	23/6/2025	25/6/2025	3 days
29. Create Data Dictionary	26/6/2025	28/6/2025	3 days
30. Develop Sequence Diagram	26/6/2025	28/6/2025	3 days
31. Develop State Transition Diagram	29/6/2025	1/7/2025	3 days
32. Develop Activity Diagram	29/6/2025	1/7/2025	3 days
33. Create Data Flow Diagram (DFD)	29/6/2025	1/7/2025	3 days
34. Design and Develop Initial Prototype using Canva	2/7/2025	2/7/2025	1 day
	Total days: 28 days		
4.0 Monitoring and controlling			
35. Test Case Design and Test Plan Preparation	3/7/2025	5/7/2025	3 days
36. Heuristic evaluation	6/7/2025	8/7/2025	3 days
37. Pilot Testing and Usability Testing	9/7/2025	13/7/2025	5 days
38. User Acceptance Testing (UAT)	14/7/2025	15/7/2025	2 days
39. Carry Out Improvements Toward the System	16/7/2025	17/7/2025	2 days
40. Control Quality	3/7/2025	23/7/2025	21 days
41. Monitor Project Work	3/7/2025	23/7/2025	21 days

42. Control Scope and Schedule	3/7/2025	23/7/2025	21 days
43. Perform Integrated Change Control	3/7/2025	23/7/2025	21 days
	Total days: 21 days		
5.0 Closing Phase			
44. Finalization of prototype	24/7/2025	25/7/2025	2 days
45. Finalization of report	26/7/2025	27/7/2025	2 days
46. Presentation to Stakeholders	28/7/2025	28/7/2025	1 day
47. User/ Stakeholder Feedback Collection	28/7/2025	28/7/2025	1 days
48. Post-Implementation Review / Lessons Learned	29/7/2025	29/7/2025	1 days
	Total days: 6 days		

Project Team		**Approval / Review Committee**	
Project Manager	Isabelle Yau Li Shuang	Sponsor	Dr Abdul Aziz Abdullah
Business Analyst	Ooi Shi Qi	Business Division Head	Tom Bray
Developer	Peggy Liow En Ki	Business Unit Head	Andy Byron
Infrastructure Engineer	Leow Yan Yi	Finance Manager	Steven Chin

3.0 Project Scheduling

3.1 Milestone

Milestones Schedule			
Milestone	**Start Date**	**End Date**	**Expected duration (days)**
Initiating phase			
1. Project Approval and Kick off Meeting	12/5/2025	12/5/2025	1
2. Identify problems in the current system and compare with other systems.	13/5/2025	13/5/2025	1
3. Identify the stakeholders and conduct stakeholder meeting.	14/5/2025	14/5/2025	1
4. Develop Project Charter	15/5/2025	15/5/2025	1
5. Identify Assumptions and Constraints	16/5/2025	17/5/2025	2
6. System Request	16/5/2025	17/5/2025	2
7. Prepare Business Case	18/5/2025	19/5/2025	2
8. Project Proposal	20/5/2025	20/5/2025	1
	Total days: 9 days		
Planning phase			
9. Background study	19/5/2025	20/5/2025	2
10. Feasibility study	21/5/2025	22/5/2025	2
11. Select Development Methodology	23/5/2025	24/5/2025	2
12. Determine Functional and Non-Functional Requirements	25/5/2025	26/5/2025	2
13. Review and Get Stakeholder Approval of Requirements	27/5/2025	27/5/2025	1
14. Develop Work Breakdown Structure (WBS)	28/5/2025	29/5/2025	2
15. Define Project Milestones and Deliverables	30/5/2025	31/5/2025	2
16. Assign Project Roles and Responsibilities (Job assignments)	1/6/2025	1/6/2025	1
17. Create Gantt Chart / Project Schedule	2/6/2025	4/6/2025	3

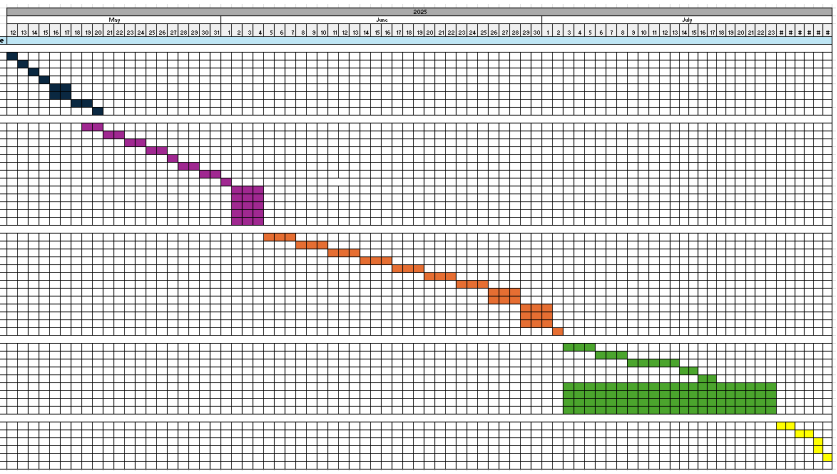
18. Develop Project Management Plan	2/6/2025	4/6/2025	3
19. Develop Risk Management Plan	2/6/2025	4/6/2025	3
20. Develop Communication Plan	2/6/2025	4/6/2025	3
21. Develop Procurement Plan	2/6/2025	4/6/2025	3
	Total days: 17 days		
Executing phase			
22. Direct and Manage Project Work	5/6/2025	7/6/2025	3
23. Acquire and Manage Project Team	8/6/2025	10/6/2025	3
24. Manage Communications	11/6/2025	13/6/2025	3
25. Integrate Planning with Execution	14/6/2025	16/6/2025	3
26. Define Backend Architecture and Data Integration Plan	17/6/2025	19/6/2025	3
27. Create Use Case Diagram and Entity Relationship Diagram (ERD)	20/6/2025	22/6/2025	3
28. Design Class Diagram	23/6/2025	25/6/2025	3
29. Create Data Dictionary	26/6/2025	28/6/2025	3
30. Develop Sequence Diagram	26/6/2025	28/6/2025	3
31. Develop State Transition Diagram	29/6/2025	1/7/2025	3
32. Develop Activity Diagram	29/6/2025	1/7/2025	3
33. Create Data Flow Diagram (DFD)	29/6/2025	1/7/2025	3
34. Design and Develop Initial Prototype using Canva	2/7/2025	2/7/2025	1
	Total days: 28 days		
5.0 Monitoring and controlling			
35. Test Case Design and Test Plan Preparation	3/7/2025	5/7/2025	3
36. Heuristic evaluation	6/7/2025	8/7/2025	3
37. Pilot Testing and Usability Testing	9/7/2025	13/7/2025	5
38. User Acceptance Testing (UAT)	14/7/2025	15/7/2025	2

39. Carry Out Improvements Toward the System	16/7/2025	17/7/2025	2
40. Control Quality	3/7/2025	23/7/2025	21
41. Monitor Project Work	3/7/2025	23/7/2025	21
42. Control Scope and Schedule	3/7/2025	23/7/2025	21
43. Perform Integrated Change Control	3/7/2025	23/7/2025	21
	Total days: 21 days		
5.1 Closing Phase			
44. Finalization of prototype	24/7/2025	25/7/2025	2
45. Finalization of report	26/7/2025	27/7/2025	1
46. Presentation to Stakeholders	28/7/2025	28/7/2025	1
47. User/ Stakeholder Feedback Collection	28/7/2025	28/7/2025	1
48. Post-Implementation Review / Lessons Learned	29/7/2025	29/7/2025	1
	Total days: 6 days		

This project has used Milestone as a tool in project scheduling. Based on Tristancho (2023), project milestone is a project planning tool that notes the start and finish of a project as well as marks the completion of a major phase of work. Project milestones help a project team to accurately estimate the time it will take to complete a project by marking important dates and events, making them essential for precise project planning and scheduling. Project managers and team members can easily track important steps in the project life cycle and better understand what needs to be done and when. In this project, milestones were clearly defined for all project phases, which includes the completion of initiating activities like project approval and project charter, the finalization of the planning phase with the approval of the requirements and project schedule, the completion of execution with the delivery of initial prototype and the sign-off of user acceptance testing in the monitoring and controlling phase. The project closure is on 29th July 2025, following the presentation to stakeholders and post-implementation review. These milestones act as checkpoints, ensuring that major deliverables are completed on time and that the project progresses as planned.

3.2 Gantt Chart

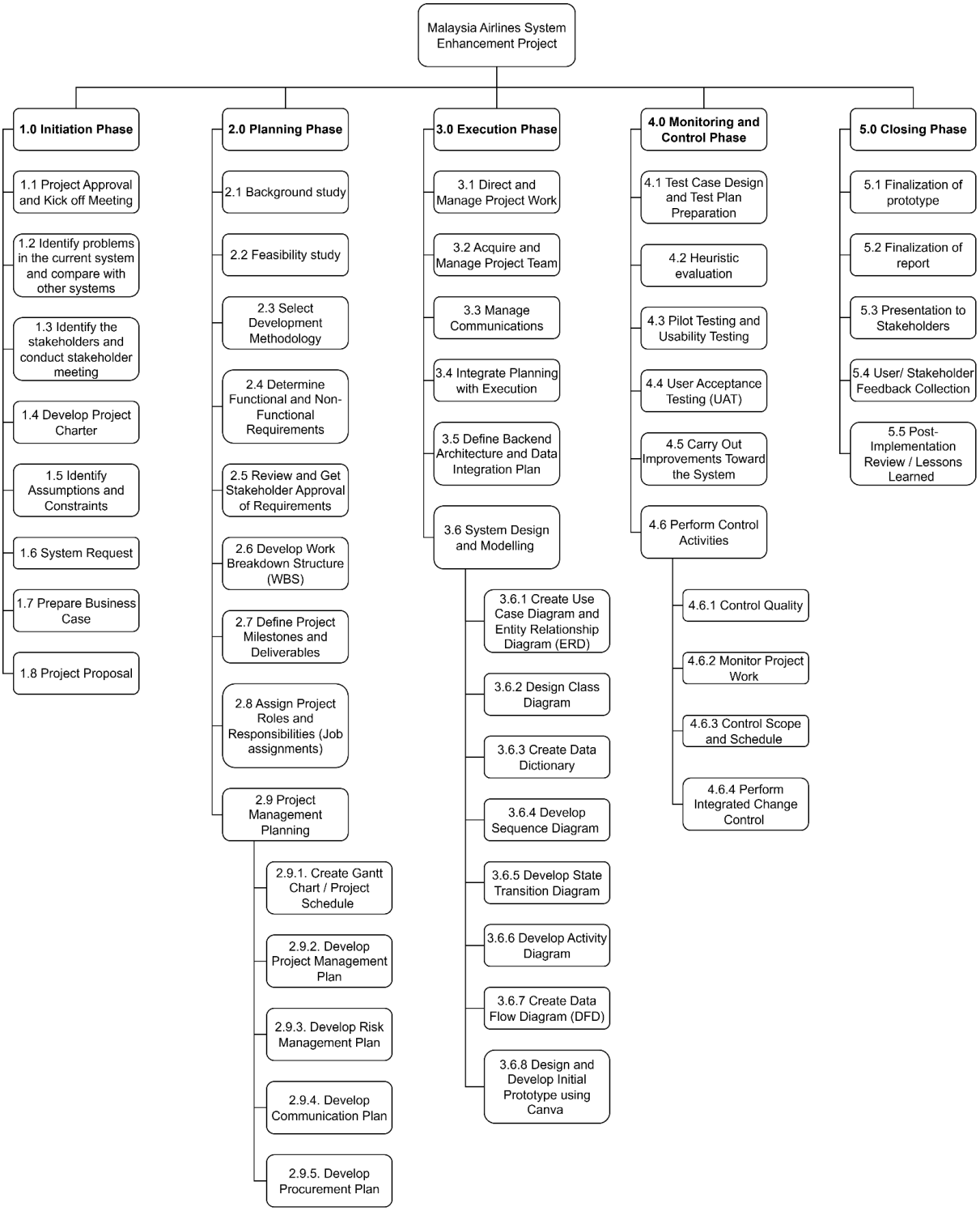
#	Activity	Start	End	Dependencies	Duration	Status	%Done
Initiation Phase							
1	Project Approval and Kick off Meeting	12/5/2025	12/5/2025		1	Completed	100%
2	Identify problems in the current system and compare with other	13/5/2025	13/5/2025	1	1	Completed	100%
3	Identify the stakeholders and conduct stakeholder meeting.	14/5/2025	14/5/2025	2	1	Completed	100%
4	Develop Project Charter	15/5/2025	15/5/2025	3	1	Completed	100%
5	Identify Assumptions and Constraints	16/5/2025	17/5/2025	4	2	Completed	100%
6	System Request	16/5/2025	17/5/2025	5	2	Completed	100%
7	Prepare Business Case	18/5/2025	19/5/2025	6	2	Completed	100%
8	Project Proposal	20/5/2025	20/5/2025	7	1	Completed	100%
Planning Phase							
9	Background study	19/5/2025	20/5/2025	8	2	Completed	100%
10	Feasibility study	21/5/2025	22/5/2025	9	2	Completed	100%
11	Select Development Methodology	23/5/2025	24/5/2025	10	2	Completed	100%
12	Determine Functional and Non-Functional Requirements	25/5/2025	26/5/2025	11	2	Completed	100%
13	Review and Get Stakeholder Approval of Requirements	27/5/2025	27/5/2025	12	1	Completed	100%
14	Develop Work Breakdown Structure (WBS)	28/5/2025	29/5/2025	13	2	Completed	100%
15	Define Project Milestones and Deliverables	30/5/2025	31/5/2025	14	2	Completed	100%
16	Assign Project Roles and Responsibilities (Job assignments)	1/6/2025	1/6/2025	15	1	Completed	100%
17	Create Gantt Chart / Project Schedule	2/6/2025	4/6/2025	16	3	Completed	100%
18	Develop Project Management Plan	2/6/2025	4/6/2025	16	3	Completed	100%
19	Develop Risk Management Plan	2/6/2025	4/6/2025	16	3	Completed	100%
20	Develop Communication Plan	2/6/2025	4/6/2025	16	3	Completed	100%
21	Develop Procurement Plan	2/6/2025	4/6/2025	16	3	Completed	100%
Execution Phase							
22	Direct and Manage Project Work	5/6/2025	7/6/2025	21	3	Completed	100%
23	Acquire and Manage Project Team	8/6/2025	10/6/2025	22	3	Completed	100%
24	Manage Communications	11/6/2025	13/6/2025	23	3	Completed	100%
25	Integrate Planning with Execution	14/6/2025	16/6/2025	24	3	Completed	100%
26	Define Backend Architecture and Data Integration Plan	17/6/2025	19/6/2025	25	3	Completed	100%
27	Create Use Case Diagram and Entity Relationship Diagram (ERD)	20/6/2025	22/6/2025	26	3	Completed	100%
28	Design Class Diagram	23/6/2025	25/6/2025	27	3	Completed	100%
29	Create Data Dictionary	26/6/2025	28/6/2025	28	3	Completed	100%
30	Develop Sequence Diagram	26/6/2025	28/6/2025	29	3	Completed	100%
31	Develop State Transition Diagram	29/6/2025	1/7/2025	29	3	Completed	100%
32	Develop Activity Diagram	29/6/2025	1/7/2025	29	3	Completed	100%
33	Create Data Flow Diagram (DFD)	29/6/2025	1/7/2025	29	3	Completed	100%
34	Design and Develop Initial Prototype using Canva	2/7/2025	2/7/2025	33	1	Completed	100%
Monitoring and Control Phase							
35	Test Case Design and Test Plan Preparation	3/7/2025	5/7/2025	34	3	Completed	100%
36	Heuristic evaluation	6/7/2025	8/7/2025	35	3	Completed	100%
37	Pilot Testing and Usability Testing	9/7/2025	13/7/2025	36	5	Completed	100%
38	User Acceptance Testing (UAT)	14/7/2025	15/7/2025	37	2	Completed	100%
39	Carry Out Improvements Toward the System	16/7/2025	17/7/2025	38	2	Completed	100%
40	Control Quality	3/7/2025	23/7/2025	35	21	Completed	100%
41	Monitor Project Work	3/7/2025	23/7/2025	35	21	Completed	100%
42	Control Scope and Schedule	3/7/2025	23/7/2025	35	21	Completed	100%
43	Perform Integrated Change Control	3/7/2025	23/7/2025	35	21	Completed	100%
Closing Phase							
44	Finalization of prototype	24/7/2025	25/7/2025	39	2	Completed	100%
45	Finalization of report	26/7/2025	27/7/2025	44	2	Completed	100%
46	Presentation to Stakeholders	28/7/2025	28/7/2025	45	1	Completed	100%
47	User Stakeholder Feedback Collection	28/7/2025	28/7/2025	46	1	Completed	100%
48	Post-Implementation Review / Lessons Learned	28/7/2025	28/7/2025	47	1	Completed	100%
Total duration					79 days		



Link to Gantt Chart: [Project management.xlsx]

Moreover, this project has also used Gantt chart as a tool in project scheduling. Solas (2025) mentioned that Gantt chart is a diagram which includes nested lists, bar charts and timelines to visually present a project's order and timing. Project managers can view the sequence of tasks, which tasks will be completed first and which ones will end later, as well as critical task dependencies, task duration and overall project duration. In this project, Gantt chart was used as a visual scheduling tool to plan, track and manage all project activities from initiation to closure. Each task was assigned with a start and end date, along with dependencies, duration and completion status. The chart helped stakeholders to visualize the overall timeline and stay informed of project updates. By organizing activities in a structured and time-bound manner, the Gantt chart plays an important role in improving coordination, managing dependencies and ensuring timely delivery of all project deliverables.

3.3 Work Breakdown Structure



Link to Work Breakdown Structure: https://drive.google.com/file/d/1WlGjExMUJvcCZOacdIcP2LlmgG4XpWuL/view?usp=sharing

Furthermore, the work breakdown structure is also used in project scheduling. Tinger (2019) said that work breakdown structure is a deliverable oriented hierarchical decomposition of the work to be executed by the project team. The project work breakdown structure identifies the major functional deliverables and subdivides those deliverables into

smaller systems and sub-deliverables. Jonas (2009) mentioned that work breakdown structure is the most important part of the project development. Without the work breakdown structure, the work schedule will be unorganized and most of the tasks will not be executable. In this project, the work breakdown structure was utilized to break down the entire scope of work into manageable components and categorized by project phases. Each major deliverable was decomposed into smaller, more actionable tasks to allow for better assignment of responsibilities, estimation of resource and progress tracking. The work breakdown structure helped the team to ensure that no essential work was overlook and that all activities are aligned with the project objectives. Through organizing the work hierarchically, the work breakdown structure provides a clear structure that improved communication, accountability and execution efficiency throughout the project lifecycle.

4.0 Project Resource

For this project, the **Responsibility Assignment Matrix (RACI Chart)** is selected as the key project resource management tool.

RACI matrix

Project Task	**Project Manager**	**Business Analyst**	**Developer**	**Infrastructure Engineer**
Define project scope and objectives	A	R		
Gather user requirements (reviews + interviews)	I	R		
Design check-in page with upcoming flights	I	C	R	C
Integrate 24-hour AI-powered live chat	C	C	R	A
Implement email-based OTP login method	I	C	R	A
Improve error message system	C	R	R	
Backend integration for chat and login features	C		C	R
Functional testing	I	R	C	C

User Acceptance Testing (UAT)	I	A	R	C
Risk and compliance (PDPA, GDPR) check	I	C		A
Project progress monitoring and updates	R		I	I
Final report and deliverable submission	A	R	C	C

R = Responsible (does the task)
A = Accountable (owns the task/deliverable)
C = Consulted (provides input)
I = Informed (kept updated)

The **RACI chart** serves as an essential tool in defining the roles and responsibilities of each team member across all project tasks. It ensures clarity in task ownership, fosters effective collaboration, and streamlines decision-making by eliminating ambiguity in who does what.

In the context of the **Malaysia Airlines System Enhancement Project**, this matrix facilitates a well-structured distribution of responsibilities among the four core team members, enabling efficient coordination, accountability, and communication throughout the project lifecycle.

- **Project Manager (Isabelle Yau Li Shuang):**
  She holds overall accountability for the project. Her key responsibilities include coordinating project activities, monitoring progress, liaising with stakeholders, and ensuring deliverables are completed within the timeline. She is accountable for initiating the project scope, tracking progress, facilitating communication, and overseeing final reporting.

- **Business Analyst (Ooi Shi Qi):**
  The business analyst is responsible for gathering user requirements through reviews and interviews, defining functional improvements, and leading User Acceptance Testing (UAT). She plays a consulted role in design and development, ensuring that user needs are accurately translated into system enhancements.

- **Developer (Peggy Liow En Ki):**
  The developer is responsible for implementing the technical features, including the redesigned check-in interface, live chat, OTP login system, and contextual error messaging. She also contributes to testing support and fixes any bugs identified during UAT.

- **Infrastructure/Security Engineer (Leow Yan Yi):**
  This role is primarily responsible for backend integration and ensuring that all new features (such as live chat and OTP) function securely and reliably. She is accountable for compliance with data protection regulations (PDPA and GDPR) and system stability during deployment.

The RACI technique is especially effective for compact teams, as it promotes transparency, minimizes role overlap, and ensures each task is assigned to the most appropriate team member. In this project, it plays a vital role in aligning responsibilities with expertise, particularly during the implementation of key enhancements like the live chat system, OTP verification, and improved user messaging.

By adopting this approach, the team maintains clear documentation of responsibilities, enabling seamless collaboration and a focused execution strategy.

5.0 Project Risk

Name	Description	Cause	Responses	Risk Owner	Probability	Impact
Incomplete Feasibility Study	May result in selection of a non-viable solution	Insufficient research or analysis during planning	Conduct a thorough feasibility study with expert input; review by stakeholders	Business analyst	Medium	High
Improper requirement gathering	Leads to future change requests or unmet stakeholder expectations	Inadequate stakeholder involvement or unclear scope	Validate requirements with stakeholders regularly	Business analyst	High	High
Poor risk, communication, or procurement issues	Delays, resource shortages, or coordination issues	Lack of attention to support plans during planning	Develop detailed and realistic plans, review and update periodically	Project manager	Medium	Medium
Inaccurate milestone or scheduling estimates	Impacts project time and cost performance	Over-optimism or lack of historical data	Use expert judgement historical data and buffer times, perform schedule validation	Project manager	Medium	High
Incomplete or incorrect system design	Late discovery of architecture flaws causes rework	Inadequate analysis, missing requirements	Conduct design reviews, prototype critical components early	Developer	Medium	High
Inadequate test planning	Bugs and errors may not be detected early	Lack of time or resources allocated for testing	Develop comprehensive test plans, schedule test phases with sufficient time	Developer	Medium	High
Low user participation in user acceptance testing	Delays in acceptance or incomplete feedback	Poor user engagement or unclear testing scope	Engage users early, send reminders	Business Analyst	High	Medium
Data from heuristic/pilot testing not actionable	Recommendation s may not be effectively implemented	Poor documentation or follow through	Assign team to analyse and convert findings into action points, track implementations	Developer	Medium	Medium

Risk register is a practical tool to support risk management. It allows the identification, analysis, and mitigation of potential risks that will hinder project progress in later phases. Our team utilised the risk register to plan ahead actions to resolve potential risks. This table includes the risk description, root causes, planned responses to the risk, person in-charge to the risks, and probability as well as impact assessment. Such detailed risk document could help in facilitating informed decision making throughout the project lifecycle as all team members could clearly understand and act upon potential risks referring to this risk document.

Eight potential risks were identified in this project. The first one being the risk of an Incomplete Feasibility Study of the issue. The insufficient study could lead to the selection of an impractical solution. This risk is identified as highly critical as it defines the project direction and its final outcomes. It is predicted that this risk will be caused by insufficient research or analysis of the situation, which needs the involvement of experts and stakeholders for review and ensurement of the technical and business feasibility.

The second risk is an Improper Requirement Gathering. This happens when stakeholder needs are not fully understood and documented. This risk is highly vital and might occur easily, which means that the project has to pay high attention to avoid the happening of such risks as this issue could lead to unmet stakeholder expectations and the need for future change requests. If this issue does happen, the business analyst must regularly validate requirements with stakeholders to prevent downstream issues like scope creep and misaligned solutions.

Poor risk, communication, or procurement planning is also another notable potential risk that highlights how inadequate support planning can trigger project delays or resource shortages. The project manager has to develop detailed and realistic plans that are regularly reviewed and updated to prevent such issues from happening.

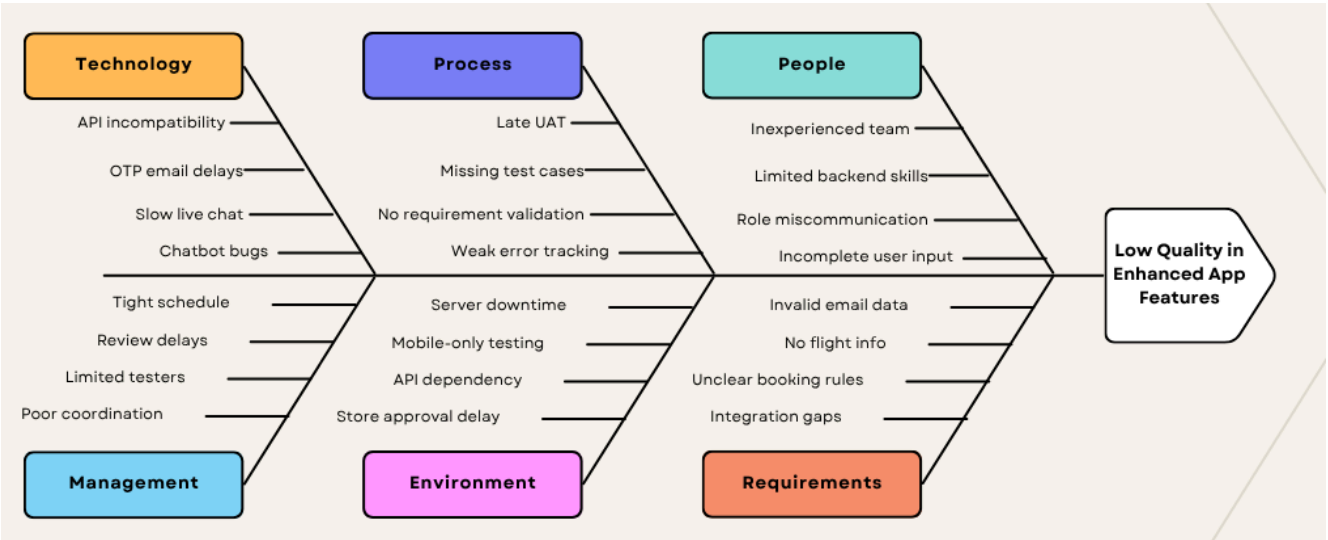
Moreover, risks such as Inaccurate Milestone or Scheduling Estimates are vital when occurred as it impacts project time and cost performance. This issue might happen due to the team's over-optimism during planning or the lack of historical data to refer to. Hence, the Project manager has to refer to expert judgement, historical data and buffer times and perform schedule validation before the executing phase.

Incomplete or incorrect system designs might also occur and lead to the late discovery of architecture flaws and causes rework of the design. This issue is caused by the inadequate analysis of the system and the missing of requirements. The developer should conduct design reviews and do prototyping of critical components early to prevent this highly impactful issue.

The risk concerning the testing phase includes the Inadequate Test Planning and Low User Participation in User Acceptance Testing are particularly important for quality assurance. Bugs that go undetected or feedback that is incomplete can threaten user satisfaction and system performance. Detailed test plans and user engagement strategies such as reminder and early involvement can be done as risk responses.

Lastly, Data from Pilot Testing Not Being Actionable is a potential risk that underscores the need to convert early test findings into meaningful insights. This issue might happen because of the poor documentation or follow through during pilot testing. This reflects that the project team should take user testing seriously and acknowledge that this process must drive actionable outcomes. Assigning a team to track these follow-ups ensures continuity between testing and implementation.

6.0 Project Quality



The Fishbone diagram has been used as our key project quality management tool for the Malaysia Airlines System Enhancement Project. It also known as an Ishikawa Diagram. The Fishbone Diagram provides a structured and visual way to identify and analyze the root causes of potential quality issues. Our specific project goal focused on enhancing the user experience of the mobile application, incorporating better check-in options, live chat 24 hours a day, 2FA via email, and deeper error messages. Early in the project, the Fishbone Diagram was a great tool in planning and managing risk in both technical and non-technical issues due to the interrelatedness of features and the ack of time with a short 11-week timeline.

The diagram helped us to identify the root causes of low-quality output through investigating the issues in six categories which are People, Process, Technology, Requirements, Environment, and Management. For the people section, we noted that the team had limited experience with backend integration and secure authentication. When considering knowledge gaps of airline APIs or chatbot systems, bugs or weak integration had a high possibility of occurring with limited knowledge. Additionally, miscommunication in allocating tasks and responsibilities clearly posed a risk that the team could not work efficiently as a team.

Other than that, in the Process section, potential sources of unresolved issues criticized the collegiality of late UAT, incomplete test cases, and limited error tracking. Hence, users may indicate a feature is working nicely. However, it required thorough and early testing to work as intended when the user engages it in a real scenario. In the Technology section, backend system incompatibility issues, email delays in receiving OTP

codes, and the chatbot not functioning were the largest concerns. The smooth integration was imperative as it impacted the reliability of the system.

Moreover, in the Management category, the schedule compressed timescales introduced challenges regarding multiple iterations, testing, and obtaining feedback. There is also a risk introduced with confusion and a lack of clarity in coordination and roles for decision making. Bringing all of these together in a Fishbone Diagram effectively focused the team on the quality goals throughout the development process.

This visual analysis gave us a platform to prioritize actions that could enable prevention, as well as uncover relationships among issues across categories. A system bug may occur in a scenario that can be traced back to poor communication or inadequate test planning. Hence, these relationships helped us to better structure our workflow with respect to deliverables and testing.

Besides, the Fishbone Diagram also guided the creation of a quality checklist specific to the project. If we identified email OTP delays under Technology, we included related test cases in the checklist to review delivery time on all scenarios. This may ensure that known risks were monitored throughout the testing. By using the diagram fostered collaboration every member of the team, which includes all of the project manager, business analyst, developer and infrastructure engineer were able to share their own perspectives. This allowed us to take a broader and more holistic view of potential quality issues thanks to the multidisciplinary involvement.

In conclusion, the Fishbone Diagram was a useful quality assurance tool. It may help us to investigate the quality issues we might run into during the lifetime of our IT enhancement project. This may include phases from the requirement gathering phase to the final testing phase. The clarity of the Fishbone diagram also helped us to identify and easily communicate quality risks to enhance decision-making and coordination. Its proactive nature is also suitable to our project since it supported the delivery of a reliable and user-friendly application, ensuring our enhancements from streamlined check-in to better security, and also met functional and user expectations.

7.0 Project Closure

For this project, the project closure technique is the Project Close-Out Meeting.

The project close-out meeting is a structured session conducted at the end of a project to formally review its performance, outcomes, and lessons learned with key stakeholders. This meeting ensures administrative closure and provides a platform to:

- Evaluate whether the project achieved its scope, time, cost, and quality objectives
- Share lessons learned and document them into the organization's knowledge base
- Review changes, challenges, and how they were addressed
- Discuss the transition plan for the enhanced Malaysia Airlines mobile app features (e.g., handing over the AI chatbot, OTP system, and new UI features to the maintenance team)

This tool is especially suitable for academic and small-scale real-world IT projects because it supports both formal closure and team reflection, and promotes knowledge transfer for future improvements.

During our close-out meeting, the team will:

- Review project deliverables (e.g., updated check-in, live chat, login, and error messaging features)
- Confirm final user acceptance
- Discuss what went well and what could be improved
- Document these outcomes into a lessons-learned summary and update process assets for future projects

This aligns with best practices outlined in the *PMBOK Guide* and ensures that our project is not only completed but formally closed with stakeholder consensus and insight capture.

7.1 Sample Project Close-Out Meeting Agenda

Project Name	Malaysia Airlines System Enhancement Project	
Meeting Type	Project Close-Out Meeting	
Date	29 July 2025	
Time	10:00 AM – 11:30 AM	
Venue	Sunway Discussion Room C	
Facilitator	Isabelle Yau Li Shuang (Project Manager)	
Attendees	Dr. Abdul Aziz Abdullah (Project Sponsor) Ooi Shi Qi (Business Analyst) Peggy Liow En Ki (Developer) Leow Yan Yi (Infrastructure Engineer)	
Time	**Agenda Item**	**Presenter**
10:00 – 10:10am	Welcome & Opening Remarks	Project Sponsor
10:10 – 10:25am	Review of Project Objectives and Deliverables	Project Manager
10:25 – 10:35am	Comparison of Planned vs Actual Schedule and Budget	Project Manager
10:35 – 10:50am	Highlights of UAT Results & Stakeholder Satisfaction	Business Analyst
10:50 – 11:05am	Discussion of Key Challenges and Lessons Learned	All Members
11:05 – 11:20am	Transition and Handover Plan (Support, Maintenance Ownership)	Infrastructure Engineer
11:20 – 11:30am	Acknowledgements & Closure	Project Manager

7.2 Lesson Learned Template

Project	Malaysia Airlines System Enhancement Project		
Team	BIS3106 April 2025		
Date Completed	29 July 2025		
Category	**What Went Well**	**What Could Be Improved**	**Recommendations for Future Projects**
Scope Management	Clear deliverables were defined early (check-in UI, live chat, email OTP)	Scope freeze came slightly late due to evolving feedback	Lock requirements earlier with sponsor confirmation
Time Management	All features delivered within the 11-week timeline	Minor delays in integration testing	Buffer at least 1 week for backend coordination
Team Collaboration	Strong communication through weekly MS Teams syncs	Slight roles overlap in UAT and documentation	Clarify overlapping responsibilities in the RACI matrix
Technical Execution	Feature enhancements worked smoothly in simulated environment	AI chatbot integration was more complex than expected	Conduct early feasibility checks and assign contingency planning
User Acceptance Testing	All scenarios were tested; received positive user feedback	Some test cases were missing for error messages	Expand testing checklist; allow more time for exploratory testing
Documentation	All mock-ups and technical notes were compiled and archived	Delay in updating version control for latest files	Use a shared cloud folder with access tracking from the start

8.0 References

Jones, E. F. (2009). Scheduling 101—the basic of best practices. Paper presented at PMI® Global Congress 2009—North America, Orlando, FL. Newtown Square, PA: Project Management Institute.

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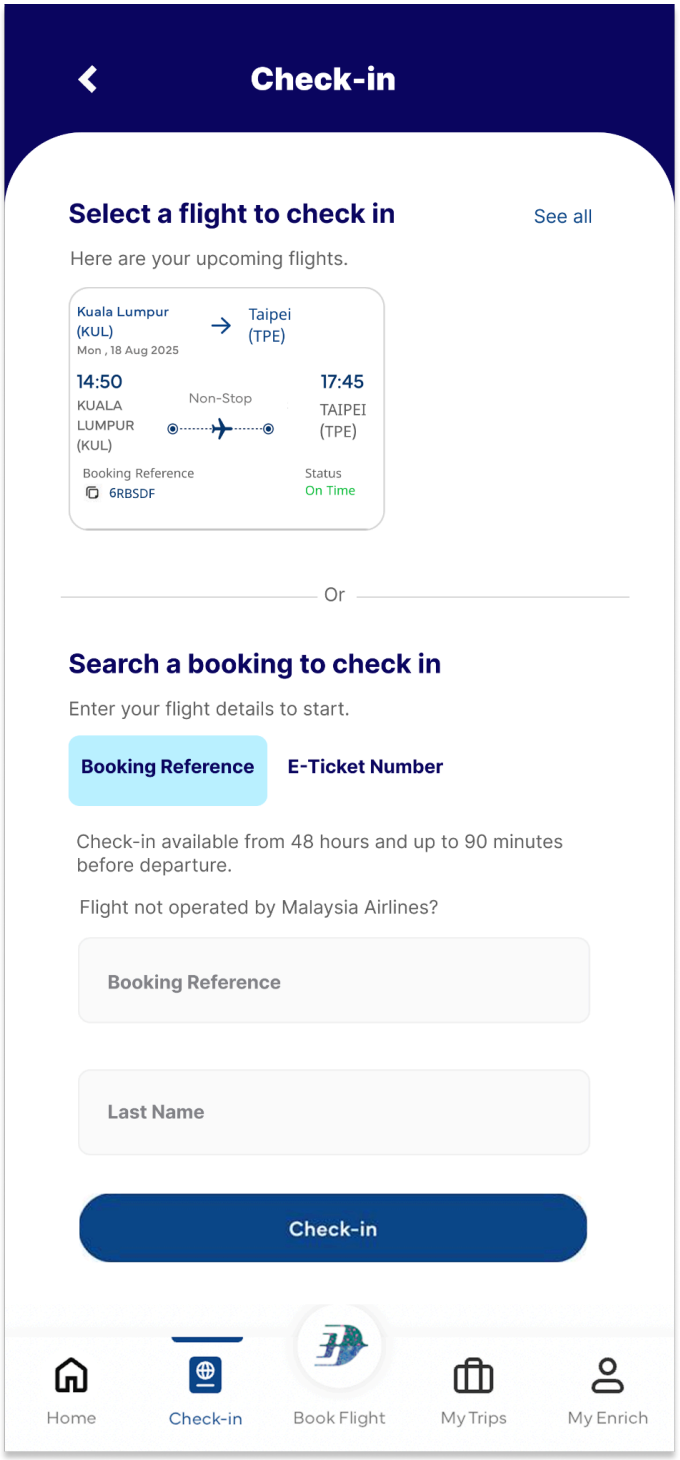
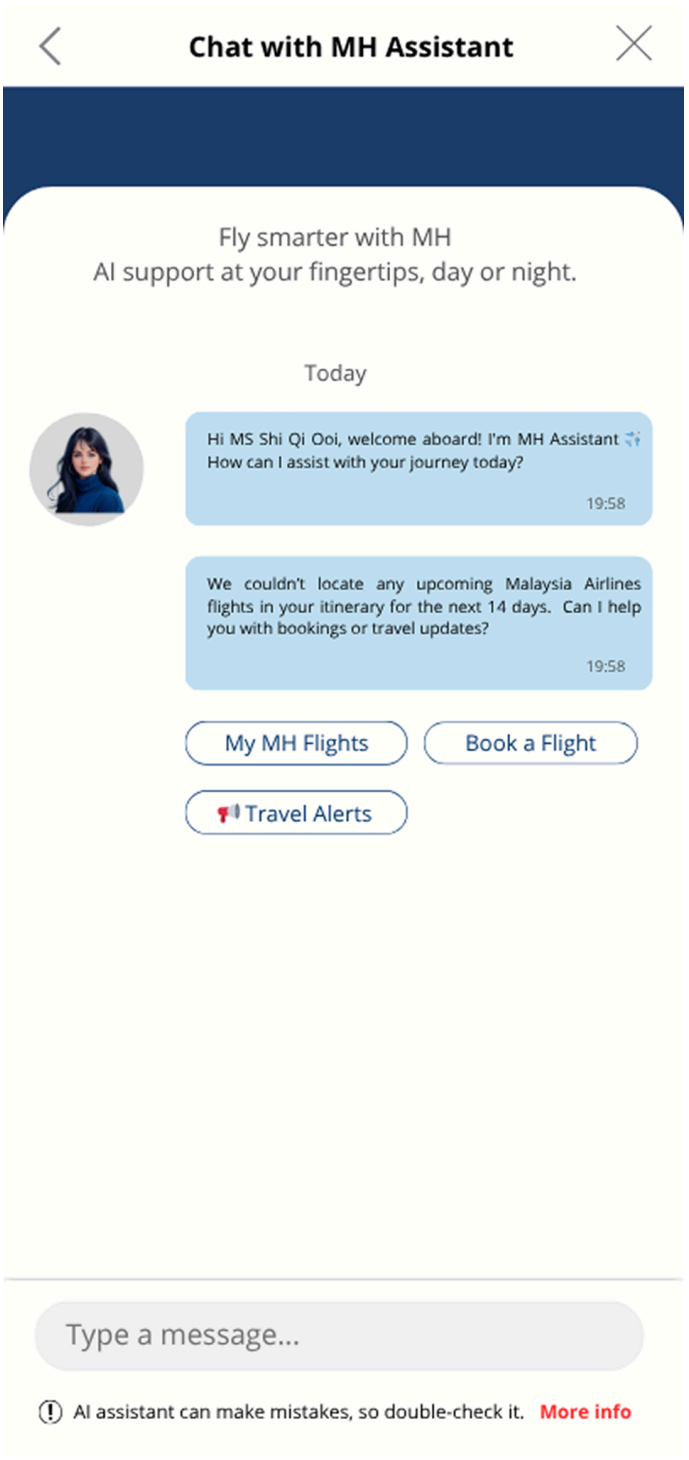
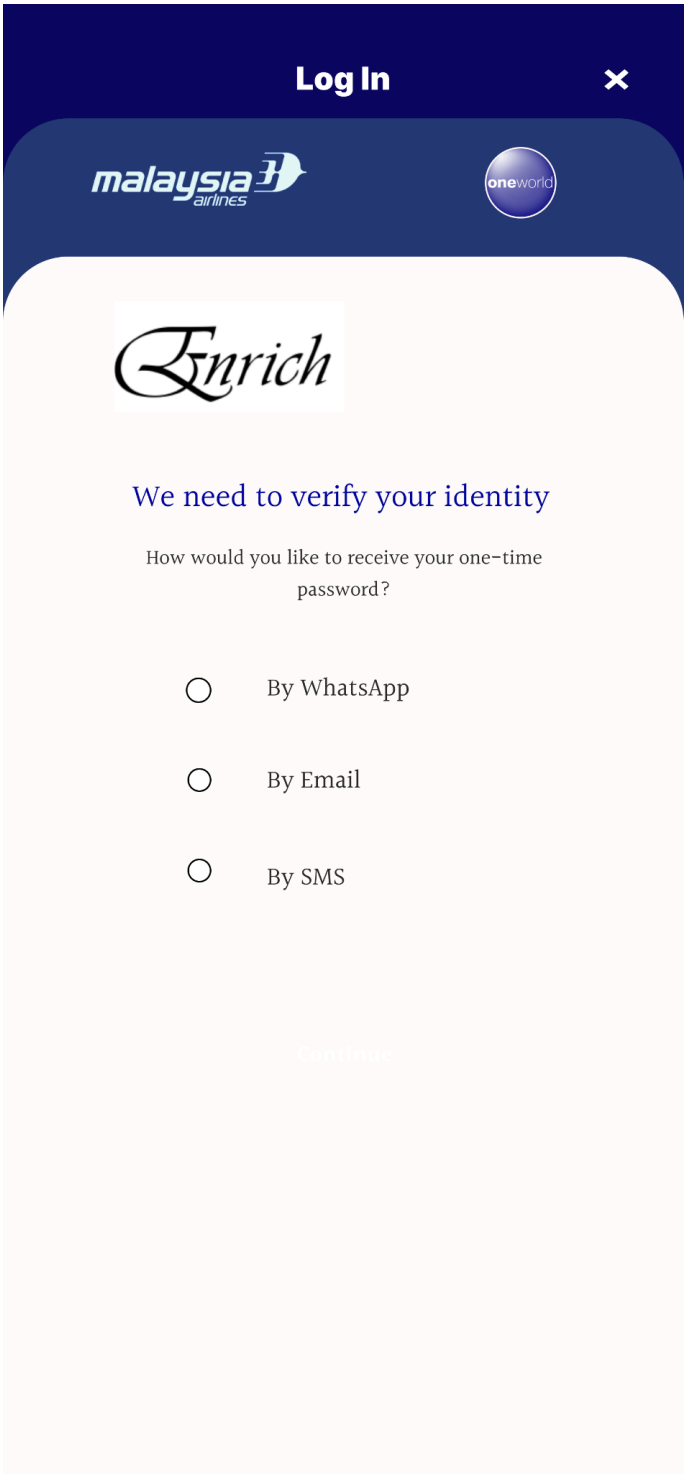
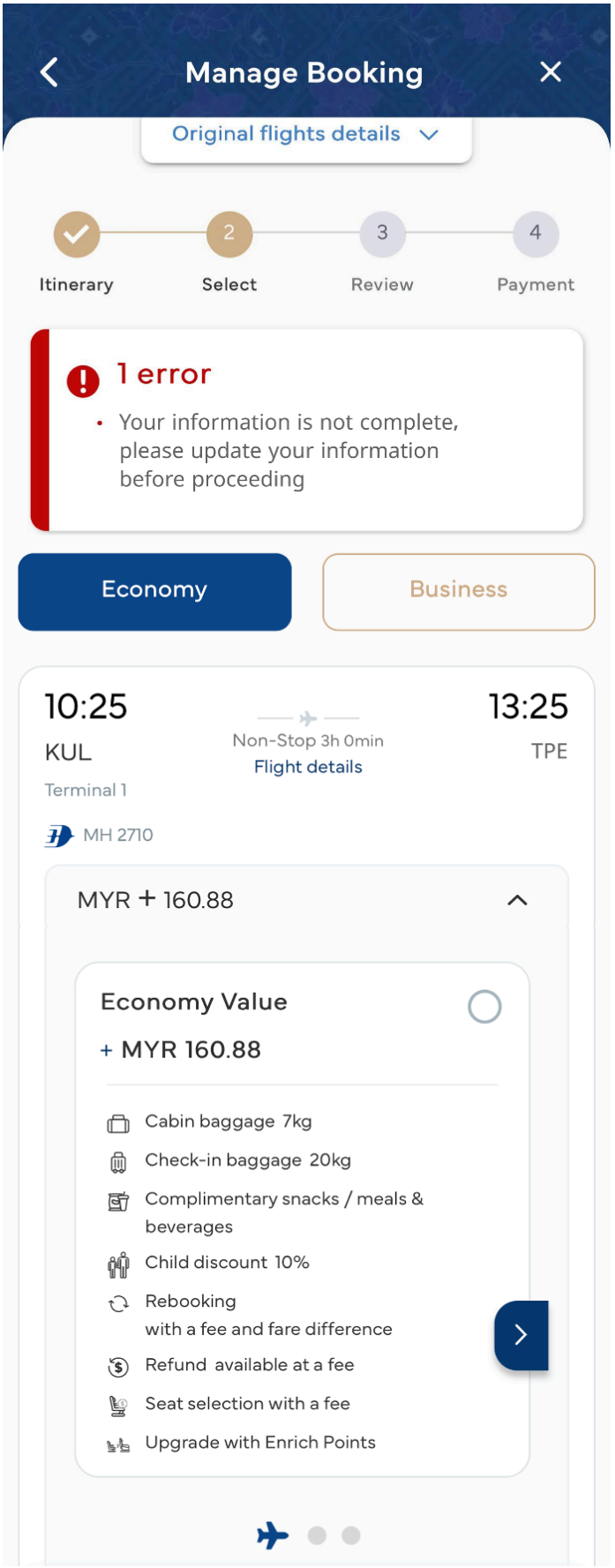
Tinger, J. (2019, Jul 2). *Work Breakdown Structure (WBS)*. ProjectManagement.com. https://www.projectmanagement.com/wikis/397395/Work-Breakdown-Structure--WBS-

Tristancho, C. (2023, April 27). *What Are Milestones in Project Management?* ProjectManager.com. https://www.projectmanager.com/blog/milestones-project-management

9.0 Appendices (Figma mockup)

1. Redesigned check-in page with added tabs of upcoming flights	2. Integrated 24-hour in-app live chat system with human agent support
	

3. Improved login experience by adding email-based OTP	4. Improved error message system with contextual error messages